Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

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Summary

CSE undergraduate at BUET focused on building intelligent systems at the intersection of large language models, software automation, and applied machine learning systems. Hands-on experience designing end-to-end AI-driven systems across mobile and backend platforms, with strong emphasis on reliability, scalability, and real-world deployment. Actively engaged in research on LLM-based software testing and medical AI, including dataset creation, evaluation pipelines, and trust-aware decision systems.

Education

Bangladesh University of Engineering and Technology (BUET) — **2022 – Present**
Bachelor of Science in Computer Science and Engineering (CSE) — **CGPA: 3.61 / 4.00**
Relevant Coursework: Data Structures and Algorithms, Digital Logic Design, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Computer Graphics, Algorithm Engineering, Artificial Intelligence, Machine Learning
Research Interests: LLM Systems, Software Testing Automation, Applied Machine Learning, Medical AI

Research Experience

- **Undergraduate Thesis: Web Testing Using Large Language Models** [GitHub]
  Python • LLMs • Web Automation • Prompt Engineering — *2025 – Present*
  - Designing LLM-based pipelines for automated web test generation using prompt-driven and programmatic approaches.
  - Creating and curating a dataset of functional descriptions and corresponding test cases for supervised evaluation.
  - Investigating coverage improvement and robustness of generated tests.
- **MedCAR: Conflict-Aware Medical Reasoning System (Under Review)** [GitHub]
  Python • Deep Learning • Medical AI — *2026 – Present*
  - Built a multi-model system for chest X-ray analysis integrating heterogeneous AI tools.
  - Developed a conflict resolution pipeline to reconcile contradictory outputs using semantic reasoning and confidence calibration.
  - Designed trust-aware decision framework with uncertainty-based abstention for reliable predictions.
- **Bengali-Loop: Long-Form Bangla ASR and Speaker Diarization Benchmark** [arXiv]
  Python • Speech Processing • Machine Learning — *2026*
  - Contributed to the development of a large-scale benchmark dataset for long-form Bangla ASR and speaker diarization.
  - Built data collection and preprocessing pipelines, including subtitle extraction and annotation for real-world speech data, enabling evaluation using Word Error Rate (WER) and Diarization Error Rate (DER).

Projects

- **MindTrace: AI-Powered Dementia Care Application** [GitHub | Feature | Infrastructure]
  Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure — *2025*
  - Engineered an AI-driven mobile–backend system for dementia care with an Android frontend and scalable Spring Boot architecture, enabling real-time caregiver workflows and seamless system integration.
  - Designed accessibility-first UI/UX for cognitively impaired users, focusing on clarity, simplified interactions, and usability to enhance engagement and reduce cognitive load.
- **Gemma VetCare: AI-Assisted Farming Support System** [GitHub | Feature]
  Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB — *2025*
  - Built an AI-assisted Android application for veterinary decision support, integrating backend services to deliver real-time recommendations for diagnosis and treatment workflows.
  - Designed RESTful APIs optimized for low-connectivity environments with efficient request handling and streamlined data flow.
- **Compiler Construction (Flex, Bison, 8086 Codegen)** [GitHub]
  C++17 • Flex • Bison/Yacc • Linux — *2024*
  - Designed and implemented a full compiler pipeline for a C-like language, encompassing lexical analysis, parsing, semantic analysis, and intermediate code generation.
  - Generated optimized 8086 assembly code with symbol table management and basic optimizations.
- **Computer Graphics Pipeline Implementation** [GitHub]
  C++17 • OpenGL/GLUT • Rasterization • Ray Tracing — *2025*
  - Implemented a complete graphics pipeline including transformations, clipping, rasterization with Z-buffering, and ray tracing.
  - Developed OpenGL-based rendering demos with camera systems supporting 6-DOF control and real-time interaction.

Technical Skills

Languages: C/C++, Python, Java, Kotlin, JavaScript, SQL
Backend: Spring Boot, Spring AI, Node.js, Express.js, REST APIs
AI/ML: PyTorch, TensorFlow, Scikit-learn, LLM Systems
Databases: PostgreSQL, MongoDB, Redis
Tools: Linux, Git, Docker, Firebase

Additional Information

Spoken Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)
Programming Profiles: NeetCode: mninadmnobo   LeetCode: mninadmnobo